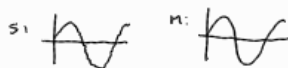
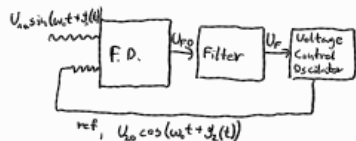


Fazeno upetna zanka (phase locked loop - PLL)



primerjano signal in modelski signal



A) Fazni detektor

$$\Delta y(t) = \Delta \omega t$$

$$U_{FD} = U_1 \cdot U_2 = U_{a0} U_{c0} \sin(\omega_c t + \phi_1) \cos(\omega_c t + \phi_2) = \frac{A}{2} U_{a0} U_{c0} [\sin(\phi_1 - \phi_2) + \sin(2\omega_c t + \phi_1 + \phi_2)]$$

$$\langle U_{FD} \rangle = \frac{A}{2} U_{a0} U_{c0} [\langle \sin(\phi_1 - \phi_2) \rangle + 0] = K_{FD} \sin \phi_c, \quad \phi_c = \phi_1 - \phi_2$$

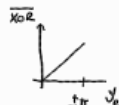
$$\phi_1 \approx \phi_2 : U_{FD} \approx K_{FD} \cdot \phi_c$$

Primer: izhodna F.O.

1) XOR vrata (ali A ali B)

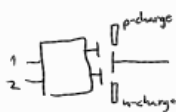


$$\rightarrow \int dt$$



- ne vemo, kateri signal prehitava/zaostaja
- če $\omega_1 = \omega_2$ pridemo do istega rezultata

2)



če prvi prehitava, na izhodu dobimo + signal

če drugi prehitava, dobimo - signal

slabost: če $\omega_1 = \omega_2$, dobimo isti rezultat

B) Filter: LPF

$$F(s) = \frac{U_F}{U_{FD}}$$

C) VCO

$$\text{izhod: } U_{c0} \cos \omega_c t = U_{c0} \cos(\omega_c t + \phi_2(t))$$

$$\omega_c(t) = \omega_0 + K_o U_F$$

$$\omega_c t = \omega_0 t + \phi_2(t) \xrightarrow{\frac{d}{dt}} \dot{\phi}_2 + \omega_0 = \omega_{n_2}$$

$$\rightarrow \dot{\phi}_2 = K_o U_F(t) \rightarrow s \phi_2(s) = K_o U_F(s)$$

$$\phi_2(s) = \frac{A}{2} K_o U_{FD} F(s) = \frac{A}{2} K_o K_{FD} F(s) \phi_c(s)$$

$$\frac{\phi_c}{\phi_1 - \phi_2} = \frac{1}{1 + \frac{K_o K_{FD} F(s)}{s}} \quad \text{Prenosna funkcija PLL}$$

Izberemo optimalno zanko $\rightarrow F(s) = ?$

obravnavamo PLL kot opt. senzor 2. reda za fazo

ϕ_2 - faza v sistemu s (ne poznamo)

$$\dot{\phi}_2 = \omega_2$$

$$\omega_2 = 0 + \omega$$

$$\hookrightarrow \text{dim. } \sin, \cos, \langle w(t)w(t') \rangle = Q \delta(t-t')$$

ϕ_1 - izmerek

$$\phi_1 = z = \phi_2 + v_2(t), \quad \langle v_2(t)v_2(t') \rangle = R \delta(t-t')$$

Kalman:

$$\begin{cases} \dot{\hat{\phi}}_2 = \hat{\omega}_2 + K_y(z - \hat{\phi}_2) \\ \hat{\omega}_2 = 0 + K_w(z - \hat{\phi}_2) \end{cases} \quad \begin{cases} \dot{\hat{\phi}}_2 = K_w(z - \hat{\phi}_2) + K_y(z - \hat{\phi}_2) / \omega \\ s^2 \hat{\phi}_2 = (K_w + s K_y)(z - \hat{\phi}_2) \\ \hat{\phi}_2 = \frac{1}{s} (K_y + \frac{1}{s} K_w)(z - \hat{\phi}_2) \end{cases}$$

$$\phi_2 = \phi_1, \quad \phi_1 = z$$

$$\phi_2(s) = \frac{1}{s} K_y K_{FD} F(s) (\phi_1 - \phi_2)$$

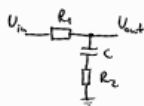
$$\rightarrow F(s) = \frac{1}{K_o K_{FD}} [K_y s + \frac{1}{2} K_w] \quad \hookrightarrow \text{integrator } H_{int}(s) = \frac{1}{s}$$

če uporabimo RC integrator ($F(s) = \frac{1}{1+\tau s}$)

$$\frac{y}{x} = \dots = \frac{s^2 + \frac{1}{\tau}s}{s^2 + \frac{1}{\tau}s + \frac{K_0 K_{FD}}{\tau}}$$

vloga τ : če daljši, daveč visoke $\omega \Rightarrow$ dušenje $\rightarrow 0$
 $\Rightarrow$ prehlajanje faze razlike $\rightarrow$ NI OK

$\rightarrow$ modificiran RC-integrator



$$\frac{U_{in}}{Z} = I = \frac{U_{in}}{R_1 + R_2 + \frac{1}{i\omega C}}$$

$$U_{out} = I Z_2 = I \left(R_2 + \frac{1}{i\omega C} \right)$$

$$F = \frac{U_{out}}{U_{in}} = \frac{1 + i\omega C R_2}{i\omega C} \frac{i\omega C}{1 + i\omega C (R_1 + R_2)} = \frac{1 + i\omega \tau_2}{1 + i\omega (\tau_1 + \tau_2)}$$

PLL:

$$\frac{y}{x} = \dots = \frac{s^2 + \frac{1}{\tau_1 + \tau_2}s}{s^2 + \frac{1 + K_0 K_{FD} \tau_2}{\tau_1 + \tau_2}s + \frac{K_0 K_{FD}}{\tau_1 + \tau_2}}$$

$$\omega_s^2 = \frac{K_0 K_{FD}}{\tau_1 + \tau_2}, \quad K_0 K_{FD} \tau_2 \gg 1$$

$$2 \xi \omega_s = \frac{1 + K_0 K_{FD} \tau_2}{\tau_1 + \tau_2} = \frac{K_0 K_{FD} \tau_2}{\tau_1 + \tau_2} = \omega_s^2 \tau_2$$

$\downarrow$
 $\frac{1}{\sqrt{2}}$

$$\Rightarrow \omega_s \tau_2 = \sqrt{2}$$

$\hookrightarrow$ določimo tako, da so enačba uveljavljena

$\Rightarrow$ PLL optimalno duši! ✓

Stabilnost PLL

$\cdot \varphi_c$ ni majhen $\Rightarrow K_{FD} \varphi_c \rightarrow K_{FD} \sin \varphi_c$

$$\ddot{\varphi}_c + \frac{1 + K_0 K_{FD} \tau_2}{\tau_1 + \tau_2} \cos \varphi_c \dot{\varphi}_c + \frac{K_0 K_{FD}}{\tau_1 + \tau_2} \sin \varphi_c = \ddot{\varphi}_n + \frac{1}{\tau_1 + \tau_2} \dot{\varphi}_n$$

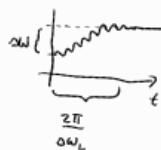
Vklapljanje PLL

$$\Delta \omega = |F(i\omega)| K_0 K_{FD}$$

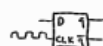
$$\Delta \omega_c = |F(i\omega_c)| K_0 K_{FD}$$

$$F(i\omega) = \frac{1 + i\Delta \omega \tau_2}{1 + i\Delta \omega (\tau_1 + \tau_2)} \approx \frac{\tau_2}{\tau_1 + \tau_2}$$

$$\rightarrow \Delta \omega_c = K_0 K_{FD} \frac{\tau_2}{\tau_1 + \tau_2}$$



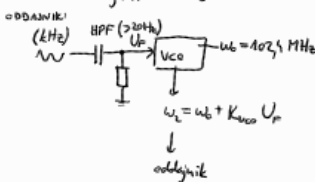
PLL primer:



(ko cirk skoči na 1, strani vrednost 0 m.g in jo dravnja)

2) Frekv. modulacija in demodulacija (FM)

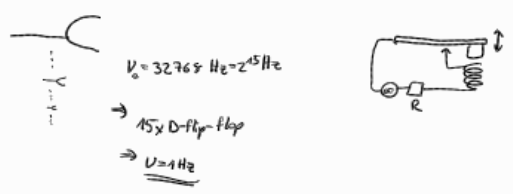
Audio signal $\sim 1 \text{ kHz}$



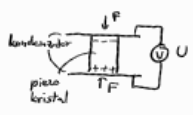
SPREJEMNIK:



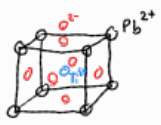
Kvarčna ura



- piezo senzor



perovskitni kristali: $(\text{SbTiO}_3, \text{SnTiO}_3)$



$$e: (2e_-) \cdot \frac{1}{8} + (-2e_-) \cdot 6 \cdot \frac{1}{2} + 4e_+ = 0$$

ob deformaciji se storiže el. + in. premenke $\Rightarrow$ el. dip. moment

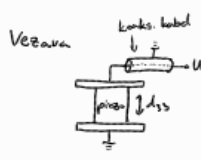
$$\langle \vec{P}_0 \rangle \neq 0 \Rightarrow \vec{P} = \sum \vec{P}_i \neq 0, \vec{P} = d \cdot \vec{T} \rightarrow \text{impetostni tenzor}$$

$$\hookrightarrow \text{piezoimp. tenzor (linearni pribl.)}$$

$$P_i = d_{ijk} T_{jk}$$

$$e_i = P S = d_{ij} \sigma_j = d_{ij} F_j$$

$$\hookrightarrow \text{piezol. tenzor}$$



izgube:
- kapacitivni tok skozi piezo
- kapacitivni tok skozi kabel
- upornost žice

$$I = I_p + I_k + I_R = (C_p U_i)' + (C_k U_i)' + \frac{U_i}{R} = d_{33} \dot{F}$$

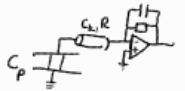
$$R d_{33} S \dot{F} = \underbrace{[(C_p + C_k) R S + 1]}_T U_i \quad \text{Hooke: } \frac{F}{S} = E \frac{\Delta x}{x_0}$$

$$R d_{33} \frac{S E}{x_0} \Delta x = (C_S + 1) U_i$$

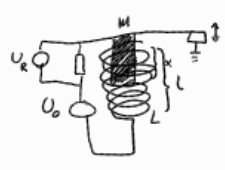
$$\frac{d_{33} S E}{(C_p + C_k) x_0} \tau_S \Delta x = (C_S + 1) U_i$$

$$\frac{U_i}{\Delta x} = k \frac{S}{\tau_S + 1}$$

problem rešimo z negativno povr. zanko OP.AMP.



- induktivni senzor



$$\phi_n = L I$$

$$\int H dl = N I$$

$$H l \approx N I$$

$$\frac{1}{2} L I^2 = W_m = \int dV w_m \approx \int dV \frac{1}{2} \mu \mu_0 \left(\frac{N I}{l} \right)^2 = \frac{1}{2} \mu_0 \left(\frac{N I}{l} \right)^2 S \left[\mu x - (l-x) \cdot 1 \right]$$

$$\rightarrow L = \mu_0 S \left[\mu x - (l-x) \right] \frac{N^2}{l}$$

$$U_i = \dot{\phi}_n = -(L \dot{I})$$

$$U_0 + U_R = \frac{\mu_0 N^2 S}{l^2} (\mu - 1) \dot{x}$$

- uporovni senzor

$$R = \rho l / S$$

$$\Delta R = \Delta l S / S + d S l / S - d S \frac{\rho}{S}$$

$$\frac{\Delta S}{S} = -2 \mu \frac{\Delta l}{l} \quad (\text{Hooke})$$

$$\frac{\Delta R}{R} = \frac{\Delta l}{l} + \frac{d \rho}{\rho} + 2 \mu \frac{\Delta l}{l}$$

$$\frac{\Delta R/R}{\Delta l/l} = k = 1 - 2 \mu + \frac{d \rho / \rho}{\Delta l / l} \approx 3$$

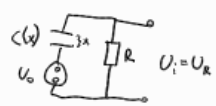
- vezava v Wheatstoneov most



$$U_m = U_0 \frac{R_2 R_3 - R_1 R_4}{(R_1 + R_2)(R_3 + R_4)} = U_0 \frac{(R_2 - R_1) R}{(R_2 + R_1) 2R} \approx U_0 \frac{\Delta R}{4R}$$

$R_1, R_2 \dots \approx 1 \text{ k}\Omega$

Kondenzatorski senzor



$$C = \frac{\epsilon \epsilon_0 S}{x}$$

$$I = \frac{dq}{dt} = \frac{d}{dt}(C U_c)$$

$$dC = -\frac{C_0}{x_0} dx$$

$$U_i R = -R C \dot{U}_i - R U_0 \left(\frac{C_0}{x_0} \right) \dot{x}$$

$$U_i (1 + \tau s) = -\tau \frac{U_0}{x_0} s x$$

$$\frac{U_i}{x} = \frac{\tau s}{1 + \tau s} \frac{U_0}{x_0}$$